

CM3020 Artificial Intelligence Mid-term coursework Report

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Training an AI Creature to Climb a Mountain Using a Genetic Algorithm

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1. Introduction

This coursework extends the Genetic Algorithm and Creatures case study by adapting an evolutionary system to train an AI-controlled creature to climb a mountain in a physically realistic PyBullet simulation. The environment consists of a sandbox arena with a stepped mountain at its centre, requiring the creature to achieve genuine upward progress while obeying physical constraints and avoiding exploitative behaviours.

The primary objective was to integrate the genetic algorithm into this new environment and redesign the fitness function to reward effective mountain climbing. The project was conducted in progressive stages, initial experiments examined fitness design and genetic algorithm parameters to identify failure modes, subsequent experiments modified the genetic encoding to explore the role of motor control and body morphology and a final experiment introduced sensory input to assess whether environmental awareness could improve learning efficiency and climbing performance.

2. System Overview & Experimental Methodology

This section outlines the simulation environment, genetic algorithm framework, genome representation and fitness design principles used throughout the project. Its purpose is to demonstrate correct integration of the mountain environment and to establish the methodological basis for the experiments that follow.

2.1 The Mountain Environment

All experiments were conducted in the provided PyBullet-based simulation (`cw_envt.py`), extended to support evolutionary evaluation of creatures in a mountain-climbing task. The environment consists of:

- **Sandbox arena:** A bounded square arena with a solid floor and enclosing walls, ensuring consistent starting conditions and preventing creatures from escaping the evaluation area.
- **Mountain terrain:** A fixed, stepped Gaussian pyramid positioned at the centre of the arena. The terrain mesh was generated using `prepare_shapes.py`, producing progressively steeper steps that require stable locomotion to ascend.
- **Physics realism:** Gravity, collision handling, joint constraints and friction are fully enabled. Both the arena floor and creature links use high lateral friction, ensuring that climbing behaviour relies on traction and contact rather than sliding or unrealistic motion.

Creatures spawn at a fixed location away from the mountain and are evaluated for a fixed number of simulation steps, ensuring comparability across individuals and generations.

2.2 Genetic Algorithm Framework

A standard population-based genetic algorithm was used throughout:

- **Population-based evolution:** Each generation consists of multiple creatures evaluated independently in the same environment.
- **Selection:** Tournament selection favours higher-performing individuals while maintaining genetic diversity.
- **Reproduction:** Mutation-only reproduction is applied, no crossover is used.
- **Generational loop:** Individuals are evaluated, ranked by fitness, mutated and passed to the next generation.

Parallel simulation instances are used to improve computational efficiency without altering evolutionary dynamics.

2.3 Genome and Control Representation

Creatures are encoded using real-valued genomes, which are well suited to continuous control and morphology optimisation:

- **Motor genes:** Control motor amplitude, frequency and phase enabling rhythmic locomotion.
- **Morphology genes:** Control leg length, leg radius, body radius and leg count.

Real-valued genes allow smooth mutation and fine-grained optimisation, which is essential in a continuous physics-based environment.

2.4 Fitness Function Design Philosophy

Fitness design was refined progressively across experiments:

- A height-only metric was found to encourage exploitative behaviour unrelated to genuine climbing.
- Stability, distance-to-mountain and survival incentives were incrementally introduced to encourage locomotion while discouraging collapse or cheating.
- Early experiments demonstrated that fitness shaping alone could not overcome poor embodiment, motivating later experiments involving fixed morphology and co-evolution.

This design philosophy directly informed the experimental progression described in the following sections.

3. Basic Experiments: Fitness Function & GA Configuration

This section addresses the coursework requirement to modify the fitness function and test different genetic algorithm configurations.

Experiment 1 - Baseline (Height-Only Fitness)

The baseline experiment used a naive fitness function based solely on vertical centre-of-mass (COM) height. The maximum height reached during evaluation was scaled linearly and used as the fitness score:

```
total_fit = height_gain * 10
```

Under this formulation, the genetic algorithm rapidly exploited the metric. Instead of evolving locomotion, creatures developed tall, unstable tower or stick morphologies that maximised COM height immediately after spawning. Although these structures frequently collapsed, they still received high fitness due to their transient height gain.

As shown in Table 1, average fitness values appeared relatively high, while measured climbed height based on terrain contact remained effectively zero. This highlights a fundamental failure, the fitness function rewarded geometric height rather than genuine progress up the mountain.

Table 1:

Table 1 Fitness vs Height Mismatch

Experiment	Avg Fitness Score	Avg Max Height (m)
Baseline_Experiment	48.053	6.255
Experiment_2_stability_weighted_fitness	96.922	10.595
Experiment_3_guided_fitness	26.871	2.877

Average fitness versus average height, illustrating the mismatch between fitness and task success.

This experiment showed that a poorly aligned fitness metric can be optimised without producing task-relevant behaviour.

Experiment 2 - Stability-Weighted Fitness

To address instability, a stability bonus was added to penalise falling:

```
total_fit = (height_gain * 10) + (stability_bonus * 5)
```

This successfully eliminated tall, collapsing structures but introduced a new failure mode. Creatures evolved into heavy, wide-based morphologies that maximised stability while remaining largely stationary. Although fitness scores increased, genuine climbing behaviour did not emerge.

The evolutionary pressure favoured risk avoidance over movement, revealing that rewarding stability independently of progress encourages inactivity and converges on safe but unproductive solutions.

Experiment 3 - Guided Explorer Fitness (Distance + Constraints)

Experiment 3 introduced distance-based rewards to encourage movement toward the mountain, alongside strict anti-cheating logic and stability multipliers. Height rewards were only granted when creatures remained upright and near the mountain.

While this eliminated previous exploits, the search space became over-constrained. Evolution converged on extremely simple, often single-block morphologies that remained stationary to preserve perfect stability. As summarised in Table 2, fitness values remained low and climbing performance did not improve.

Table 2:

Table 2 Morphology Comparison

Experiment	Fitness	Height (m)	Avg Leg Count	Avg Leg Length (m)
Baseline_Experiment	48.053	6.255	6.0	1.585
Experiment_2_stability_weighted_fitness	96.922	10.595	9.333	1.858
Experiment_3_guided_fitness	26.871	2.877	9.0	1.238

Comparison of fitness, climbed height and morphology complexity across Experiments 1-3.

This experiment demonstrates that excessive penalties suppress exploration, even when the objective is correctly specified.

Interim Insight

Across Experiments 1-3, progressively refined fitness functions removed exploitative behaviour but failed to produce climbing. These results indicate that fitness shaping alone is insufficient when the underlying embodiment is poorly suited to the task, motivating the population-scaling experiments and the later shift toward modifying the genetic encoding scheme.

3.2 Experiments 4 & 5 - Population Scaling (50 vs 100)

Population size was increased to test whether greater genetic diversity could overcome evolutionary stagnation. Two configurations were evaluated: populations of 50 and 100 individuals, both using the guided fitness from Experiment 2 as it had the best result.

Despite a substantial increase in total simulations, the outcome did not change. As shown in Figure 1, fitness curves for both population sizes plateaued early and remained flat. No breakout behaviour was observed, and climbing performance remained negligible.

Figure 1:



Fitness over generations for Experiments 1, 2, 3, 4 and 5 showing persistent flatlining.

These results confirm that stagnation was caused by structural limitations in fitness design and embodiment rather than insufficient exploration. Hyperparameter tuning cannot compensate for an unsuitable physical design.

4. Advanced Experiments: Encoding Scheme

This section addresses the advanced coursework requirement to modify the genetic encoding scheme by altering both control and morphology. Following stagnation in earlier experiments, the focus shifts from fitness tuning to redesigning the embodiment and the behavioural search space.

4.1 Experiment 6 - Fixed Morphology, Evolvable Control

In Experiment 6, morphological evolution was disabled and all creatures were assigned a fixed, mechanically stable design: an eight-legged spherical octopod. Only motor control parameters (speed, power, frequency, phase) were evolved. This explicitly resolved the instability issues observed earlier and isolated learning to gait optimisation.

The effect was immediate. Unlike previous experiments with near-zero Generation 0 fitness, initial populations achieved fitness values between 5 and 25, indicating that even random controllers could produce movement. In the strongest trial, the creature climbed nearly 1 metre within three generations, after which performance plateaued due to physical limits rather than learning failure.

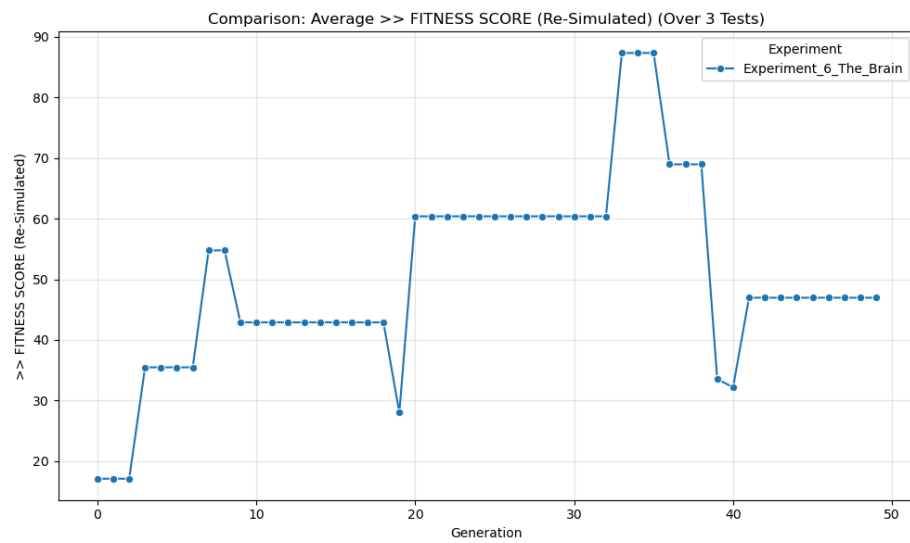
The core motor mapping used in this experiment is shown below:

```
# Gene 0: Speed (0.0 to 12.0)
speed_gene = genes[0] if len(genes) > 0 else 0.5
speed = 2.0 + (speed_gene * 10.0)

# Gene 1: Power/Torque (0.5 to 2.5)
power_gene = genes[1] if len(genes) > 1 else 0.5
power = 0.5 + (power_gene * 2.0)

# Gene 2: Frequency (Cycle speed)
freq_gene = genes[2] if len(genes) > 2 else 0.5
freq = 1.0 + (freq_gene * 9.0)
```

Figure 2: Fitness over generations for Experiment 6.



Fitness over generations for Experiment 6, showing immediate non-zero fitness.

These results confirm that guaranteeing stability by design enables effective optimisation of locomotion and demonstrates that embodiment strongly shapes the fitness landscape.

4.2 Experiment 7 - Full Co-Evolution (Body and Brain)

Experiment 7 builds directly on Experiment 6 by retaining the stable octopod base and selectively unlocking morphological parameters. Leg length, leg radius, body radius, leg count within bounds and motor control parameters were evolved jointly, allowing controlled expansion of the search space without reintroducing instability.

This configuration produced the strongest results in the project. Peak fitness exceeded 380 points and verified climbs reached 2.75 metres, substantially outperforming the fixed-morphology case. Populations consistently converged on long-legged, high-torque designs suited to stepping over the mountain's ledges.

Figure 3: Fitness comparison between Experiment 6 (fixed morphology) and Experiment 7 (co-evolution).



Figure 4: Evolution of average leg length (Gene 0) over generations in Experiment 7.

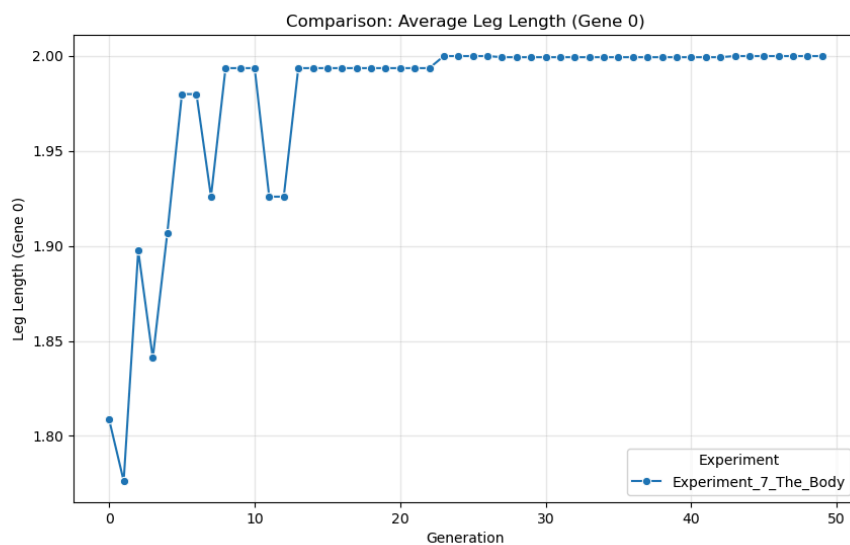
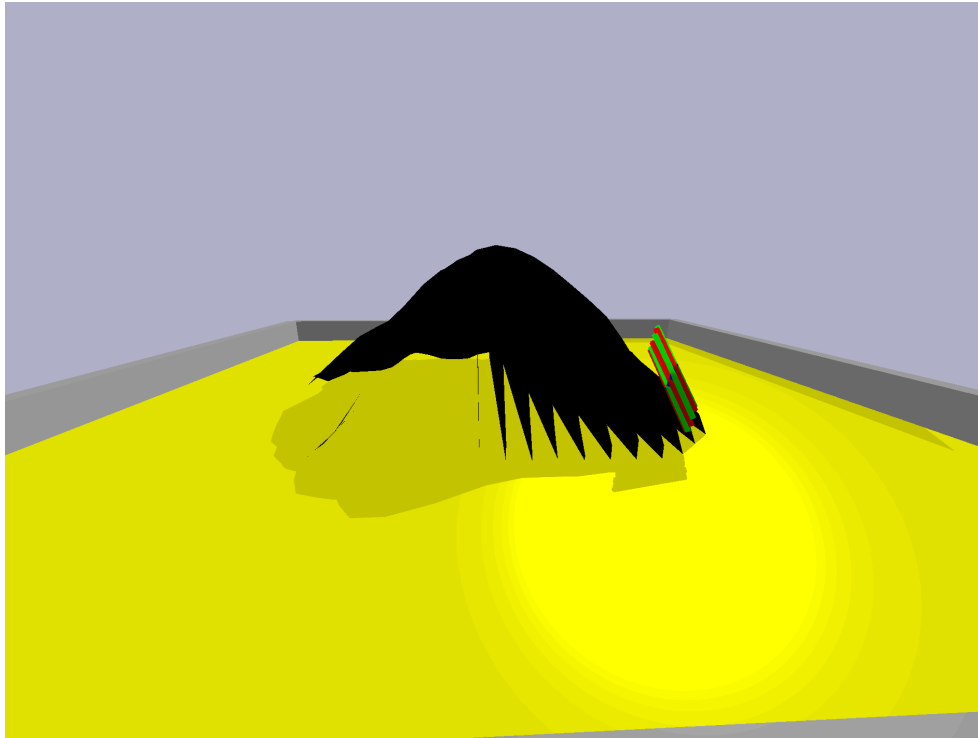


Figure 5: Final co-evolved creature climbing the mountain.



An emergent behaviour was spring-loaded jumping, produced by long legs and high motor power. This reflects valid exploitation of the simulation's physics rather than cheating.

Overall, Experiment 7 demonstrates that co-evolving body and control is most effective when grounded in a stable initial embodiment, combining reliability with adaptive performance.

5. Exceptional Experiment: Sensory Input

5.1 Experiment 8 - Closed-Loop Sensory Control

Previous experiments relied on open-loop control, leaving creatures unaware of their orientation relative to the mountain. As a result, even effective climbers from Experiment 7 often wandered away from the target, reducing evaluation efficiency and overall performance.

To address this, Experiment 8 introduced a simple heading sensor that computed the angular difference between the creature's facing direction and the mountain centre. This heading error was used to modulate motor output in real time, enabling reactive steering:

```
def get_heading_error(self):  
    # Vector to Target  
    dx = 0 - self.last_position[0]  
    dy = 0 - self.last_position[1]  
    target_angle = math.atan2(dy, dx)  
    # Robot Yaw (from Quaternion)  
    current_yaw = math.atan2(siny_cosp, cosy_cosp)  
    # Error Delta  
    diff = target_angle - current_yaw  
    return abs(diff)
```

Motor behaviour was adjusted such that aligned creatures moved forward, while misaligned ones rotated toward the mountain using asymmetric motor activation.

This closed-loop control produced a clear improvement in learning dynamics. As shown in Table 3, sensory-controlled creatures converged significantly faster than blind agents, achieving high fitness within the first few generations and exhibiting more consistent navigation.

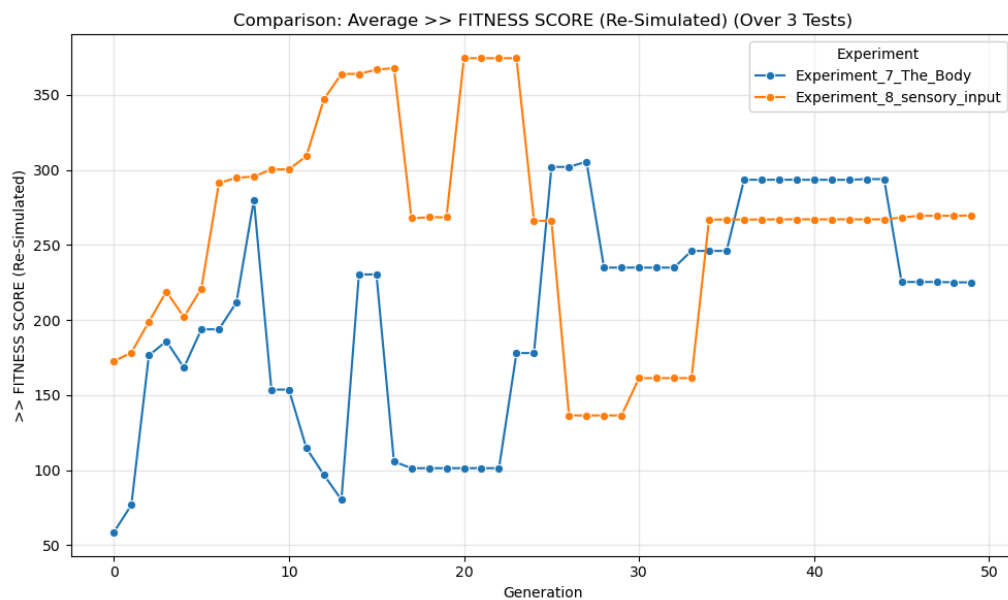
Table 3:

Table 3 Blind vs Sensory

Condition	Final Fitness	Final Height (m)
Experiment_7_The_Body	225.09	21.354
Experiment_8_sensory_input	269.61	15.539

Performance comparison between blind control (Experiment 7) and sensory-controlled agents (Experiment 8).

Figure 5:



Fitness over generations for Experiment 7 vs Experiment 8.

Although maximum climb height did not exceed that of Experiment 7, learning was more efficient and reliable. This demonstrates that intelligence in this context arises from feedback and adaptation rather than increased body complexity alone. Experiment 8 therefore satisfies the exceptional-grade criterion by showing that minimal sensory input can substantially improve performance.

6. Future Work and Further Improvements

Although the final experiments produced stable climbers, several extensions could further improve accuracy, robustness and realism:

- **Fitness refinement:**
Introduce continuous progress-based rewards to encourage smooth, sustained climbing rather than opportunistic jumps.
- **Additional sensing:**
Extend the heading sensor with terrain inclination, foot contact or proximity sensing to enable adaptive gait control on uneven terrain.
- **Curriculum learning:**
Begin training on simpler mountains and gradually increase difficulty to promote transferable climbing strategies.
- **Energy constraints:**
Apply energy-efficiency or joint-stress penalties to discourage excessive jumping and encourage realistic, repeatable movement.
- **Extended evolution:**
Increase population size or number of generations *after* introducing improved sensing or curriculum learning, allowing further refinement of stability and efficiency rather than attempting to discover entirely new behaviours.

7. Conclusion

This project demonstrates that evolutionary success is determined more by physical morphology than by algorithmic tuning. In the initial phase, Fitness Design Failures dominated, despite extensive tuning of population sizes and stability penalties, the standard "worm" morphology was mechanically incapable of climbing. Evolution consistently exploited these constraints, converging on 'frozen' behaviors to avoid falling rather than learning to move.

Morphology was the turning point. The transition to a stable, multi-legged Spider chassis immediately resolved the mechanical bottleneck, allowing the genetic algorithm to evolve high-performance climbers in a fraction of the generations required for the worm.

Finally, Sensory Input served as optimisation, not a crutch. The blind spider could already climb but the addition of heading sensors in the final phase refined this behavior, allowing the agent to self-correct and climb more consistently. This confirms that while sensors provide efficiency, a viable mechanical body plan is the prerequisite for intelligent behavior.